

# XAI: LMs for Commonsense Reasoning & Contrastive Explanations

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**eXplainable**  
Artificial Intelligence

## Disclaimer ---

This course is based on

### **Explainable Artificial Intelligence**

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

# Explain Yourself!

## Leveraging Language Models for Commonsense Reasoning

Rajani, McCann, Xiong & Socher

Presenters: Karly Hou, Eshika Saxena, Leonard Tang, Kat Zhang

## Introduction ---

- **Commonsense reasoning:** making human-like presumptions and judgements about ordinary situations
- Modern ML methods struggle with commonsense reasoning
- Explanations help verbalize reasoning that models learn while training
- Common sense Question Answering (CQA) dataset

### Example

**Question:** While eating a *hamburger with friends*, what are people trying to do?

**Choices:** **have fun**, tasty, or indigestion

**How do these models perform reasoning and to what extent is that reasoning based on world knowledge?**

## 🍷 Key Contributions: CoS-E Dataset ---

Common Sense Explanations (CoS-E): Collected human explanations (annotations and natural language explanations) to build on top of CQA



## Key Contributions ---

- 1 Common Sense Explanations (CoS-E)
- 2 Commonsense Auto-Generated Explanations (CAGE)
- 3 CAGE outperforms best baseline by 10% and produces explanations to justify its predictions
- 4 Explanation transfer on two out-of-domain datasets

## Related Work: Commonsense Reasoning ---

- Commonsense reasoning datasets:
  - ▶ Story Cloze: predicting story ending from a set of plausible endings
  - ▶ Situations with Adversarial Generations (SWAG): predicting next scene based on initial event
- Models achieve human-level performance on some datasets
- Models struggle with understanding how pronouns resolve between sentences and world knowledge
- CQA addresses this by requiring models to infer from the question
- Language models perform poorly compared to human participants on CQA

**Unclear: Do models actually do common-sense reasoning?**

## 🍷 Related Work: Natural Language Explanations ---

- Rationale generation by highlighting complete phrases in input text that are sufficient to predict desired output (Lei, Barzilay, and Jaakkola 2016)
- Human-generated natural language explanations to train a semantic parser to generate noisy labeled data and train a classifier for generating explanations (Hancock et al. 2018)
- Interpretability comes at the cost of loss in performance on Stanford Natural Language Inference dataset (Camburu et al. 2018)
- Multi-modal: Ensemble explanations and visual explanations improve performance (Park et al. 2018)

**Do explanations for CQA lead to improved performance?**

## Related Work: Knowledge Transfer in NLP ---

- Reliance on transfer of knowledge through pre-trained word vectors (e.g. Word2vec, GloVe) and contextualized word vectors (more refined with general encoding)
- Language models trained from scratch on large amounts of data and fine-tuned on specific tasks perform well
  - ▶ Only a few parameters need to be learned from scratch
  - ▶ Perform well on small amounts of supervised data
- **Gap: Fine-tuned language models don't perform as well on CQA**

**Can we leverage these models to generate explanations and show that these explanations capture common sense?**

# Common Sense Explanations (CoS-E)

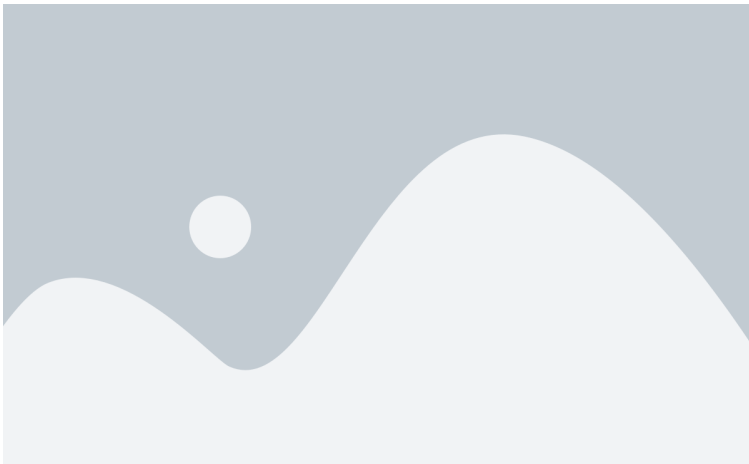
## 🍷 Dataset Structure/Creation ---

- Based on the CQA dataset
- CoS-E provides natural-language explanations for the correct answer choice and highlights important words in the question
- Explanations generated using MTurk
- Goal: To show whether models are performing reasoning correctly



## 🍷 Dataset Considerations ---

- *CoS-E-selected* refers to the highlighted words, *CoS-E-open-ended* refers to the explanations
- Quality control was performed on the annotations and explanations
- Even explanations that don't discuss the ground truth answer are useful



# Commonsense Auto-Generated Explanations (CAGE)

## 🍷 CAGE Phase 1 ---

CAGE Phase 1:

- Provide CQA example alongside corresponding CoS-E explanation to a language model
- Train model to generate the CoS-E explanation



## 🍷 CAGE Phase 2 ---

### CAGE Phase 2:

- Use language models to generate explanations for each example in the training and validation sets of CQA
- Provide CAGE explanations to a second model by concatenating it to the original input (question, answer choices, and language model output)



## 🍷 CAGE Phase 1: Intuition ---



## How is CAGE Trained? ---

- Language Model trained to generate explanations from question-answer choice pairs
- Use pretrained OpenAI GPT
- Fine-tuned on the CQA and CoS-E dataset combination
- Two possible settings:
  - ▶ **explain-then-predict (reasoning)**
  - ▶ **predict-then-explain (rationalization)**

## CAGE Notation ---

- Question  $q$
- Answer choices  $c_0, c_1, c_2$
- Correct answer  $a \in \{c_0, c_1, c_2\}$
- CoS-E explanation  $e_h$
- CAGE predicted explanation  $e$

## Reasoning ---

- Model is fine-tuned on the question, answer choices, and explanation tokens, but **not** the actual label.

$$C_{RE} = "q, c_0, c_1, \text{ or } c_2? \text{ commonsense says } "$$

- Objective Function (canonical conditional language modeling objective):

$$\sum_i \log P(e_i \mid e_{i-k}, \dots, e_{i-1}, C_{RE}; \Theta)$$

## Rationalization ---

- Model is now also given the ground truth label  $a$ :

$$C_{RA} = "q, c_0, c_1, \text{ or } c_2? a \text{ because } "$$

- Objective Function is the same as before but also conditioned on the *label*  $a$
- Thus, the explanations create rationalization that makes the model more interpretable

## Training Parameters ---

- Generate sequences of maximum length 20
- Batch Size: 36, Epochs: 10
- Selected the best model using BLEU and perplexity scores

# Commonsense Predictions with Explanations

## 🍷 CAGE Phase 2: Intuition ---



## CAGE Inference ---

- Given human explanation from CoS-E or LM reasoning, can then perform predictions on CQA
- Simply concatenate Question, [Sep], Explanation, [Sep], Answer Choice as input to downstream CSRM (classifier)
- Use binary classification head on top of BERT backbone
  - ▶ 3 answer choices → 3 input sequences
  - ▶ Take sequence yielding highest confidence as output

## 🍷 CSRM (BERT) Training Hyperparameters ---

- Train batch size: 24
- Test batch size: 12
- 10 training epochs
- Max sequence length of 50 for labels-only; 175 including explanations

# Experimental Results

## 🍷 Experimental Results: CQA with CoS-E ---



Table 2: Results on CQA dev-random-split with CoS-E used during training.

## 🍷 Experimental Results: Comparison with SOTA ---

- Google search “question + answer choice” collected 100 top snippets per answer as context for **Reading Comprehension model**
- Extra data did not improve accuracy
- CAGE-reasoning resulted in **10% accuracy gain** over previous SOTA



## 🍷 Experimental Results: Oracle Upper-Bound ---



Table 4: Oracle results on CQA dev-random-split.

- Oracle upper-bound: **human-generated explanations** from CoS-E provided during training and validation
- Unfair setting because human had ground truth answer
- “CoS-E selected”: explanation consists of words humans selected as justification for model

## 🍷 Transferring Explanations Across Domains ---

- How well do natural language explanations transfer from CQA to SWAG and Story Cloze Test?
- Use GPT CAGE model fine-tuned on CQA train/dev to generate explanations on SWAG and Story Cloze Spring 2016 train/val
- Rinse and repeat using BERT with classifier head

## 🍷 Experimental Results: Domain Transfer ---



- Camburu et al (2018): transferring explanations from SNLI to MultiNLI performs very poorly
- Transfer of explanations on commonsense reasoning tasks
- NLI has small fixed set of pre-defined labels unlike commonsense reasoning tasks (CQA, SWAG, Story Cloze)
- **Adding explanations led to very small decrease in performance**

# Qualitative Analysis

## Analysis of CAGE ---

- CAGE-reasoning at train + validation  $\rightarrow$  72% accuracy
- CoS-E-open-ended performance at 90%—why the gap?
- Measure quality of CAGE:
  - ▶ Human evaluation (42% CAGE vs 52% CoS-E-open-ended)
  - ▶ BLEU score measures syntactical precision by n-gram overlap
  - ▶ Perplexity: token-level measure of how well language models predict next word
- Result: beneficial to fine-tune the LM, but humans and LMs have widely varying ways of providing useful explanations

## 🍷 Analysis of Baseline BERT Model ---

- Error analysis on baseline BERT w/o explanations: performs poorly on longer/more compositional questions → explanations help
- CAGE reasoning typically simpler construction than CoS-E-open-ended, but adds meaningful context
- However, CAGE still provides “incorrect” answers often



## 🍷 Domain Transfer: SWAG + Story Cloze ---



## Conclusion & Group Discussion



## Conclusion ---

- CoS-E on top of CommonsenseQA
- CAGE framework → LM leverages explanations
- Classifier on top of explanations
- SOTA performance on a difficult commonsense reasoning task
- Opens further avenues for studying explanation as it relates to interpretable commonsense reasoning

## Discussion Questions ---

- BLEU and perplexity as measures of goodness?
  - ▶ Has been shown multiple times to correlate poorly with human judgement
- Joint training of explanation and label? Not one prior to the other
- Can commonsense reasoning help general reasoning (e.g. mathematics, Fermi, counterfactual) in other domains?
- How to align human/machine explanations?
- Recent work (RLPrompt, AutoPrompt) has shown that optimal prompts for LMs are often gibberish
  - ▶ What does this say about the validity of using SOTA LMs for explanations?
  - ▶ Can we regularize LM training to better align with human reasoning?

# Interpreting Language Models with Contrastive Explanations

Kayo Yin & Graham Neubig

Presented by Charumathi Badrinath, Eric Shen, Leonard Tang, and Skyler Wu

## Motivation + Example ---

- We've seen many interpretability and explanation strategies being applied to LMs, including transformer-based autoregressive LMs. . .
  - ▶ **Gradient-based/erasure-based feature attribution** methods provide a straightforward way to do this
  - ▶ Interpret each token of the input text as a feature
  - ▶ At each step, use gradients to calculate a **saliency score** to quantify the importance of each previous token to the model output (i.e. logit prediction for the next token)
- E.g.  $\text{gradient} \times \text{input}$

## 🍷 Motivation + Example: How It Works ---



## 🍷 Motivation + Example: The Problem ---

- **Problem:** For many LMs, using typical gradient- or erasure-based methods doesn't provide informative explanations
- Most of the time, the token with the highest saliency is the token immediately before the prediction
- How can we create more meaningful attributions? By *contrasting* them with other token predictions.



## 🍷 Main Contributions and Key Ideas ---

- ① **Contrastive explanations:** why did the model predict one token *instead* of another?  
Extended previous methods.
- ② **Grammatical consistency:** contrastive  $>$  non-contrastive explanations w.r.t. verifying linguistic/grammatical phenomena.
- ③ **Human simulatability:** contrastive explanations help users better predict LLM behavior, also found to be more useful by humans.



## Background: GPT-2 ---

- Authors focus on GPT-2 (1.5B) and GPT-Neo (2.7B) → very similar to each other
- **Training:** WebText dataset of 8 million web-pages
  - ▶ No task-specific supervised training = “Multi-task training”
- **Objective:** predict the next word, given all previous words in input
- **Behavior:** “chameleon-like,” adapts to style + content of the input text
- **Architecture:** Transformer-based
  - ▶ “Autoregressive”: outputs tokens one at a time, *but* each token generated is appended to the input sequence → fed back to model for next step

## Background: Gradient Norm Saliency Scores ---

- **Originally for image classification:** compute gradient of class score w.r.t. input image, take the norm.
  - ▶ Big gradient = big influence.
- **For LLMs:** compute gradient of next token logit w.r.t. current input.

$$g(x_i) = \nabla_{x_i} q(y_t | \mathbf{x})$$

$$S_{GN}(x_i) = \|g(x_i)\|_{L_1}$$

(Simonyan, Vedaldi, and Zisserman 2013)

## Background: Gradient $\times$ Input Saliency Scores ---

- **Method:** Similar gradient computation as Gradient Norm, simply replacing  $L_1$  norm with dot product with input itself.

$$g(x_i) = \nabla_{x_i} q(y_t \mid \mathbf{x})$$

$$S_{GI}(x_i) = g(x_i) \cdot x_i$$

(Shrikumar et al. 2016)

## 🍷 Background: Input Erasure Saliency Scores ---

- **Intuition:** how does erasing different parts of the input affect the output?
- **Procedure:** compute difference in model outputs using full input vs. input with a specific token zeroed out. NOT gradient-based!

$$S_E(x_i) = q(y_t \mid \mathbf{x}) - q(y_t \mid \mathbf{x}_{\neg i})$$

(Li et al. 2016)



## Background: Related Work + Limitations (Pt. 1) ---

- Non-contrastive saliency score methods:
  - ▶ Simonyan et al. 2013, Shrikumar et al. 2016, Li et al. 2016 — not NLP specific!
  - ▶ Not very developed in NLP use cases.
  - ▶ When applied to NLP (Wallace et al. 2019), methods often return last token before output as most influential.
- Adversarial Methods on NLP:
  - ▶ Wallace et al. 2019: HotFlip on NLPs, replace words to change model's prediction. AllenNLP suite.

## 🍷 Background: Related Work + Limitations (Pt. 2) ---

- Counterfactual explanations in *text classification*:
  - ▶ Jacovi et al. 2021: erase features, project into “contrastive space” → measure importance by comparing class probabilities before/after erasure.
  - ▶ Unsure how to extend into *language modeling* space with much bigger input + output spaces.
- Contrastive methods are not new, just not used for NLP very much (Stepin et al. 2021, survey).

## 🍷 Method: Contrastive Setup ---

- Simple modification to formulation of existing gradient-based explanations.
- **Contrastive** setup:



## 🍷 Method: Contrastive Saliency Formulas ---

Let  $q(y_t|\mathbf{x})$  be the model output for token  $y_t$ ,  $S(x_i)$  the saliency score for token  $x_i$  in input  $\mathbf{x}$ , and  $\mathbf{x}_{-i}$  the input  $\mathbf{x}$  where  $x_i$  is zeroed out.



## 🍷 Method: Back-Propagation in Contrastive Setting ---



## Evaluation: Grammatical Consistency ---

**Q:** Are contrastive explanations  $\gg$  non-contrastive in identifying words that we think should influence the output token?

### Experimental Setup

- **BLiMP dataset:** pairs of minimally different English sentences that contrast in grammatical acceptability under some linguistic paradigm
- 5 linguistic phenomena with 12 paradigms
- Used spaCy NLP library to extract grammatically relevant parts of each sentence

# 🍷 Evaluation: BLiMP Linguistic Paradigms ---



## Evaluation: Linguistic Consistency Metrics ---

- $S$  = explanation vector;  $S_i$  = saliency of  $x_i$
- $E$  = known evidence;  $E_i = \mathbf{1}(x_i \text{ grammatically influences output token})$
- $S \cdot E \rightarrow$  sum of saliency scores of all input tokens that are part of evidence
- **Probes needed**  $\rightarrow$  ranking of first token  $x_i$  where  $E_i = 1$  when sorted by decreasing saliency
- **MRR (mean reciprocal rank)**  $\rightarrow$  average (over all sentences) of inverse rank of first  $x_i$  where  $E_i = 1$  sorted by descending saliency

## 🍷 Findings: Linguistic Agreement ---

- Contrastive explanations are more aligned with linguistic paradigms
- Contrastive explanations have a better alignment with BLiMP than random vectors baseline
- Non-contrastive explanations do **not** outperform random baseline



## 🍷 Findings: Linguistic Agreement (Distance Effect) ---

- Further apart *known evidence* token is from *target token* → larger increase in MRR alignment of contrastive cf. non-contrastive
- Contrastive explanations can particularly capture model decisions requiring *longer-range context*



## 🍷 Evaluation: Human Simulatability ---

**Q:** Do contrastive explanations increase users' ability to predict a model's output token (i.e. "simulate" model behavior)?

### Experimental Setup

- Model = GPT-2; explanation = {no explanation,  $S_{GI}$ ,  $S_{GI}^*$ ,  $S_E$ ,  $S_E^*$ }
- 10 word pairs from BLiMP, 10 word pairs selected to maximize confusion score on WikiText-103 test split



## 🍷 Findings: User Alignment ---

- All four types of explanations help users simulate model behavior
- *Contrastive explanations* lead to *more accurate simulations*
- *Contrastive explanations* are considered *more useful*
- Takeaway: contrastive explanations help human observers accurately simulate model predictions the most



## Evaluation: What Context Do Models Use? ---

**Q:** How do language models achieve various linguistic distinctions? Is similar evidence necessary to disambiguate foils that are similar linguistically?

### Experimental Setup

- Targets = 10 most frequent words per major POS; Foils = 10000 most frequent vocab items
- For each target  $y_t$  select 500 sentences from WikiText-103
- For each foil  $y_f$  generate contrastive explanation  $e(x_i, y_t, y_f)$  and concatenate
- Apply k-means on explanation vectors  $e(x_i, y_t, \text{all foils})$

### Idea

Explanation vectors represent *type of context* needed to disambiguate foil from target

## 🍷 Findings: What Context Do Models Use? ---



# Strengths / Weaknesses ---

## Strengths

- Simple yet effective modification applicable to a variety of feature attribution methods
- Easy to compute, extensible to general LMs (including NMT)
- Evaluated with interesting foil clustering analysis
- Empirically shown to help with human observers (good for interpretability)

## Weaknesses

- Only applied to three feature attribution methods in the paper
- Only GPT-2 and GPT-Neo used as LM examples
- Human study very limited in scope
- Does not attempt to look at model internals; saliency scores are arguably a crude approximation of interpretation

## Questions for the Audience ---

- Given that LLMs often exhibit “phase shifts” at different sizes, to what extent do you expect the results to generalize to cutting-edge models like GPT-4?
- In practice, how would one create foils for free-response questions and/or general conversational use? How generalizable are these contrastive tools?
- How effective do you think saliency scores (through gradient/erasure-based methods) are for achieving interpretability?
- How much do we trust the GPT-2 embeddings (primary workhorse for most methods) and the generalizability of the authors’ results?

# Thank You!

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